

Sukruta Prakash Midigeshi

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Education

Indian Institute of Technology, Gandhinagar
B.tech in Computer Science and Engineering

Nov, 2020 – June, 2024
CGPA: 9.07/10

Publications

Achieving Limited Adaptivity for Multinomial Logistic Bandits

SP Midigeshi, T Goyal, G Sinha | RLC'25

Plan-RAG: Planning-guided Retrieval Augmented Generation

P Verma, SP Midigeshi, G Sinha, A Solin, N Natarajan, A Sharma | Workshop on Reasoning and Planning for Large Language Models@ICLR'25

Characterizing deep research: A benchmark and formal definition

Under Review

A Java, A Khandelwal, SP Midigeshi, A Halfaker, A Deshpande, N Goyal, A Gupta, N Natarajan, A Sharma | SEA@NeurIPS '25

Batched Contextual Slate Bandits with GLM Rewards

Under Review

T Goyal, SP Midigeshi, G Sinha

Work Experience

Research Fellow, Microsoft Research

July, 2024 - Present

Manager - Dr.Gaurav Sinha

- Developed a state-of-the-art Retrieval-Augmented Generation (RAG) framework, Plan*RAG, which constructs Directed Acyclic Graphs (DAGs) to effectively answer queries. Outperformed leading RAG frameworks such as Self-RAG and ReACT, and achieved a workshop publication at ICLR.
- Developed efficient algorithms for logistic and multinomial contextual bandits (a subset of Reinforcement Learning) under limited adaptivity constraints. Demonstrated both empirical and theoretical efficiency, and extended the approach to large action spaces for successful production deployment.

Software Intern, D.E.Shaw

March 2023 - June 2023

Advisor - Mr.Devender Lega

- Created various metrics for evaluating LLMs based on structured and unstructured datasets and investigated the working of BERT, RoBERTa and other Open Source models.
- Developed a recursive AI-based architecture on top of TAPAS, a table parsing model for answering questions based on custom structured data. This was done in order to combat the lack of dataset present for training a model accurately.
- Improved programming and design skills, gained knowledge of AI and machine learning techniques.

Software Intern, eLitmus

January - June 2022

Advisor - Mr.Shireesh Jayashetty

- Worked on a face recognition software and a source code plagiarism detector building upon standard checkers like MOSS and JPlag.
- The plagiarism checker first created a parse tree out of the code, using Lark which takes the grammar of the language as an input. Now an N-gram model was used to compare the similarities. Thus the checker detected plagiarism across languages, name changes as well as line switches.

Software Intern, goGlocal

June 2022 - July 2022

Advisor - Mr.Shubham Kedia

- Automated HS(Harmonized System) code generation based on user images and descriptions using ML.
- Gained experience in web development, web scraping, and ML.

Featured Projects and Collaborations

Approximating Neural Tangent Kernels using LSH

September '23

w/ Prof. Anirban Dasgupta

- Explored methodologies for approximating the Neural Tangent Kernel (NTK). Studies the used of Locality Sensitive Hashing (LSH) for various approximations, focusing on its use in kernel ridge regression.
- Since LSH is generally used for shift-invariant kernels, we tried various methodologies of expanding it to NTKs. Proved the theoretical bounds for various techniques and made many experimental observations to narrow our approach to the best estimate.

Envy-Free Budget-Feasible Allocations

August '22

w/ Prof. Neeldhara Misra

- Studied and analysed algorithms that produce EF1 or α -EF1 allocations for the fair division problem. Came up with an exact EF-1 algorithm for the 2-agent problem with identical valuations belonging to the polynomial complexity class.
- Theorized the time complexity of various algorithms centered around envy free allocations like ExistsUMandEF1 which finds out if an EF1 allocation can exist within a UM allocation.

Study on properties of undirected power graphs and their partitions.

September '23

w/ Prof. Bireswar Das

- Studied the properties of power graphs of finite abelian groups, including the partition of the graphs into triangles and their clustering coefficient.
- Came up with an algorithm for partitioning a subset of abelian groups into triangular segments. Introduced a new efficient formula for global clustering coefficient of power graphs of abelian groups. Also found an algorithm for the global clustering coefficient for cyclic abelian groups.

Teaching Experience

Teaching Assistant, Machine Learning

January '24 - May '24

Prof. Nipun Batra

- Making of the teaching, tutorial and testing material for the course ES 335(Machine Learning).
- Conducting tutorials for students along with doubt clarification sessions.

ADH (Academic Discussion Hours) Mentor, Theory of Computing

January '24 – May '24

Prof. Balagopal Komarath

- Conducting doubt clarification sessions for the students in order for them to better understand the course.

Teaching Assistant, Discrete Mathematics

Jan '23 - April '23

Prof. Bireswar Das

- Helped in the organization and structuring of the course - Discrete Mathematics offered at IIT GN.
- Facilitated engaging discussions, assisted with problem-solving sessions, and provided guidance to students.

Achievement AND Awards

- Received the Dean's List for excellence in academics for the semesters 2, 5, 6, 7 and 8.
- Earned the highest grade (11/10) in Natural Language Processing course for exceptional performance in a class of ~100 students.

Relevant Courses

- Discrete Mathematics [A], Machine Learning [A], Compilers [A-], Complexity Theory [A], Advanced Algorithms [A-], Data Science [A], Natural Language Processing [A+], Theory of Computing [A-], Combinatorics with Applications in Computer Science [A], Deep Learning [A]

Leadership and Extracurricular Achievements

- **Design Coordinator, Blithchron '21 (Cultural Fest - IITGN)** Helped lead the design team of Blithchron, the cultural fest of IIT GN. Coordinated between all the departments of the organising committee and managed the team.
- **Second Seed, Inter-IIT'22 Squash Team** — Represented IIT Gandhinagar as a member of the squash team at the prestigious Inter-IIT Sports Meet 2022.